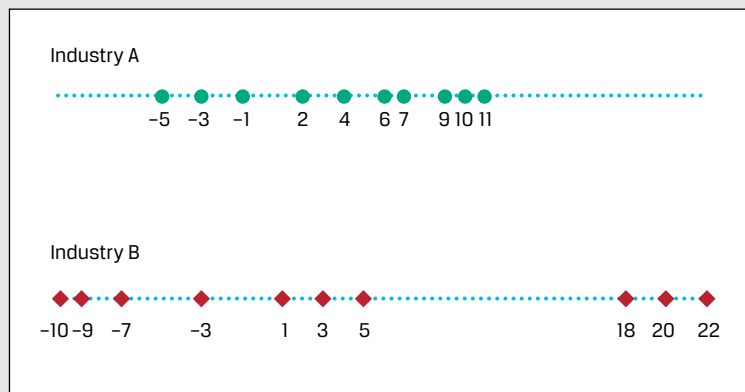


Exhibit 13: Returns on Assets Depicted Graphically

1. Calculate the average ROA for each industry.

Solution:

The arithmetic mean ROA for both industries is the sum of the 10 observations, which in both cases is 40, divided by the 10 observations, or $40/10 = 4\%$.

2. Calculate the standard deviation of ROA for each industry.

Solution:

Using Equation 5, the standard deviation for Industry A is 5.60 and 12.12 for Industry B.

3. Calculate the coefficient of variation (CV) of the ROA for each industry.

Solution:

Using Equation 7, the CV for Industry A = $5.60/4 = 1.40$ and the CV for Industry B = $12.12/4 = 3.03$.

Though the two industries have the same arithmetic mean ROA, the dispersion is quite different—with Industry B's returns on assets being much more disperse than those of Industry A. The CV for these two industries reflects this, with Industry B having a larger CV. The interpretation is that the risk per unit of mean return is more than two times ($2.16 = 3.03/1.40$) greater for Industry B than Industry A.

QUESTION SET

Consider the annual MSCI World Index total returns for a 10-year period, as shown in Exhibit 14:

Exhibit 14: MSCI World Index Total Returns

Year 1	15.25%	Year 6	30.79%
Year 2	10.02%	Year 7	12.34%
Year 3	20.65%	Year 8	-5.02%